Technical Proposal: PV + BESS for LLC “Deep Protein Processing”

We propose a 3.6 MWp solar power plant and a Huawei grid-forming energy storage system of 7.5 MW / 15.05 MWh for the plant, connected at 35 kV. The battery and the solar plant each have their own transformer station: the BESS is on a JUPITER-9000K-H1, the PV plant is on a separate STS-3000K. The project costs about EUR 3.88 million and pays back in 3.4 years. The NPV at 10% is EUR 4.96 million and the IRR is 28.0%.

Three containers mean that the 5 MW critical load stays covered even if one container is out of service. A single SPPC2000-MGCC controller runs the plant: it keeps grid import under 15 MW, forms an island when the grid is lost and sheds non-critical feeders so that the critical load stays supplied. Implementation takes about 12 months.

1. Summary and baseline data

The client’s connection limit is 15 MW, the load peak reaches 20 MW, the critical load is 5 MW, and a 5 MW CHP unit already runs on site. We install three LUNA2000-5015-2S containers (3 × 2.5 MW). Each container is an independent block, so the loss of any one still leaves 5.0 MW for the critical load, while the CHP remains the main source in normal operation. Revenue comes from day-ahead market arbitrage and from savings on grid tariffs for PV energy; on top of that the BESS keeps import under 15 MW and gives an island reserve.

The main parameters of the solution are summarised below.

Parameter	Value
PV plant	3.6 MW (DC) / 3.0 MW (AC), 10 × SUN2000-330KTL-H1, own STS-3000K
BESS power	7.5 MW (3 × 2.5 MW, 36 × LUNA2000-213KTL-H0)
BESS capacity	15.05 MWh nominal (3 × 5.015), ≈13.5 MWh usable above the 10% reserve
C-rate	0.5 C (2.5 MW per 5.015 MWh)
Critical-load security	N-1: any two containers give 5.0 MW
Fenced BESS area	≈607 m² (29.6 × 20.5 m), PV field excluded
CAPEX	≈EUR 3.88 million (with 20% contingency and optimiser software)
First-year net benefit	≈EUR 1 157 thousand
Payback	3.4 years base, 4.4 years stress (BESS margin × 0.7); IRR 28.0%, NPV at 10% EUR 4.96 million
Lifetime	8 200 cycles / 20 years (Huawei, declaration under EU Reg. 2023/1542)

Table 1. Baseline parameters of the solution

PV generation is 1 154.6 kWh/kW according to the PVGIS-5 report for the Cherkasy region, which gives about 4 157 MWh per year.

1.1 Key design decisions: why this and not that

No.	Decision	Alternatives considered	Why we chose it
1	3 × LUNA2000-5015-2S, 7.5 MW / 15.05 MWh	2 containers; 4-6 containers (20-30 MWh)	Smallest size that keeps 5 MW with one container out and also covers reactive power; best payback and IRR; 4+ need a second BESS transformer and lean on the price-taker assumption
2	One STS-9000K for all 36 PCS	Two STS-6000K; STS-9000K + STS-3000K for the BESS	7.5 MW fits 9 MVA (8.5 MVA at long-term PCS maximum); one 35 kV bay instead of two; EUR 153 thousand instead of 248 thousand
3	PV on its own STS-3000K	PV inverters on the BESS STS; PV through a new 6 kV transformer	PV inverters must not share the BESS transformer in a microgrid; STS-3000K is the smallest Huawei unit that fits 3.0 MW AC and reports to the same SACU / SPMS
4	Only SPPC2000-MGCC as plant controller	SPPC + SPPC2000-MGCC	It is the same controller family; MGCC alone gives island, black start and load shedding; EUR 36 thousand saved
5	Own LP optimiser as third-party EMS	Fixed windows only; commercial EMS platform	MGCC does not take market orders; the optimiser adds ≈EUR 48 thousand/year for EUR 30 thousand + 10 thousand/year; commercial platform prices were not found
6	Day-ahead arbitrage + PV self-consumption	Intraday trading; ancillary services	Taken from official websites
7	Dynamic SOC reserve 10% / 30%	Fixed 30%; no reserve	Each 10% costs ≈EUR 90 thousand/year; the CHP is the main backup, so 30% is held only when the CHP is out
8	Two 6 kV bus sections with bus-section breaker QS; shedding of non-critical	Separate critical bus with new switchgear; island the	Island sources (≈10-15 MW) cannot carry 20 MW; shedding

No.	Decision	Alternatives considered	Why we chose it
	feeders by the MGCC	whole plant	uses existing breakers
9	Fire wall on the north side only	Walls on all sides; no wall	Required only where the 30.5 m distance to workshops is not met; other sides meet the manual without a wall
10	PV 3.6 MWp / 3.0 MW AC	Larger PV	Fits the STS-3000K exactly; a larger plant needs a bigger transformer and more land, whose availability we do not know
11	All equipment in one phase	Two containers now, third later	Three containers are needed from day one for the critical load; one mobilisation and one commissioning
12	One cycle per day	Two cycles per day	8 200 cycles over 20 years is ≈1.1 cycles/day; a second daily spread is small in the 2026 prices

Table 2. Key design decisions and rejected alternatives

2. Equipment specification

The Huawei equipment costs EUR 2 119 100 at DAP Ukraine pricest. With the PV field, works, contingency and the optimiser software the total CAPEX is about EUR 3.88 million.

No.	Item	Qty	Price, EUR	Total, EUR
1	LUNA2000-5015-2S container (5 015 kWh, 2 507.5 kW, liquid cooling, LFP)	3	490 000	1 470 000
2	LUNA2000-213KTL-H0 PCS (12 per container)	36	6 600	237 600
3	JUPITER-9000K-H1 STS (9 MVA, 35 kV), BESS only	1	153 000	153 000
4	STS-3000K (3 MVA, 35 kV), PV only	1	105 000	105 000
5	SUN2000-330KTL-H1 PV inverter	10	6 700	67 000
6	SPPC2000-MGCC (plant and microgrid controller)	1	20 000	20 000
7	SPMS (local station server, HMI)	1	54 000	54 000
8	DTS (auxiliary transformer)	1	12 500	12 500
	Huawei subtotal			2 119 100
9	PV modules and mounting structures, 3.6 MW			≈900 000
10	35 kV bay for the BESS, installation, civil works for 3			≈148 000

No.	Item	Qty	Price, EUR	Total, EUR
	containers, fire wall			
11	Second 35 kV bay and foundation for the PV STS-3000K			≈25 000
12	Load shedding: motor operators / trip coils on 3 feeders and the bus-section breaker QS, telemetry I/O to the MGCC			≈15 000
13	Contingency 20%			≈641 400
14	Optimiser software			30 000
	Total CAPEX			≈3.88 million

Table 3. Equipment specification and CAPEX breakdown

Three containers are needed because the 5 MW critical load must survive the loss of one block: two containers give exactly 5.0 MW with no margin, and with one container out only 2.5 MW would remain. The 36 PCS load the STS-9000K with at most 7.5 MW, inside the rated 9 MVA. The SPPC is not purchased separately, because the SPPC2000-MGCC already performs the plant-control functions.

2.1 Power check with reactive power

Two containers give 5.0 MW only if the load runs at a power factor of 1. A real industrial load also draws reactive power, so the PCS must deliver apparent power. We take a power factor of 0.9 for the critical load. Then 5 MW needs 5.56 MVA, including 2.42 Mvar of reactive power.

Each LUNA2000-213KTL-H0 delivers 213 kW rated at 40 °C (192 kW at 50 °C) and up to 236.4 kVA in the long term (170.6 A at 800 V), with a power factor from 1 leading to 1 lagging. The site maximum is +35 °C, so there is no temperature derating.

Configuration	PCS	Apparent power, rated / long-term max, MVA	Active power from batteries, MW	5 MW at PF 0.9 (5.56 MVA)
2 containers	24	5.11 / 5.67	5.0	No: below the requirement at rated power, nothing left if one container fails
3 containers	36	7.67 / 8.51	7.5	Yes, with 38% margin at rated power
3 containers, one out	24	5.11 / 5.67	5.0	Yes only at the long-term maximum (2% margin) or with the CHP running

Table 4. Critical-load check with reactive power

Three containers therefore cover the critical load with active and reactive power. With one container out, they still cover it together with the CHP, which stays the main source. Losing both the CHP and a container at the same time is a double fault; in that case the MGCC may shed part of the critical load. On the transformer side, 36 PCS at their long-term maximum give 8.51 MVA, still under the 9 MVA of the STS-9000K.

3. Single-line diagram, battery connection and layout

3.1 Connection of the BESS and the PV plant

The BESS and the PV plant are connected to the 35 kV busbar through two separate transformer stations. The STS-9000K carries only battery PCS: its LV winding A takes BESS1 and racks 1-3 of BESS2 (18 PCS, 3.83 MW), and LV winding B takes racks 4-6 of BESS2 and BESS3 (18 PCS, 3.83 MW). The PV plant has its own STS-3000K with ten SUN2000-330KTL-H1 inverters (10 × 300 kW = 3.0 MW AC), so PV inverters never connect to the battery transformer.

The LV side of the STS-9000K has 30 outgoing breakers, 15 per winding, and each breaker takes one or two PCS. We connect the two PCS of one battery rack to one breaker: QA1-QA9 on winding A and QB1-QB9 on winding B. That leaves 6 spare breakers per winding, and the DTS takes one breaker on winding A. One breaker per rack also means that a single rack can be switched off without touching the others.

Inside each container there are 6 battery racks of 8 packs. Each rack goes through its RCM to two PCS, so one container feeds 12 PCS. Each rack with its two PCS has its own STS LV breaker, so a faulty rack is isolated alone while the other five keep working.

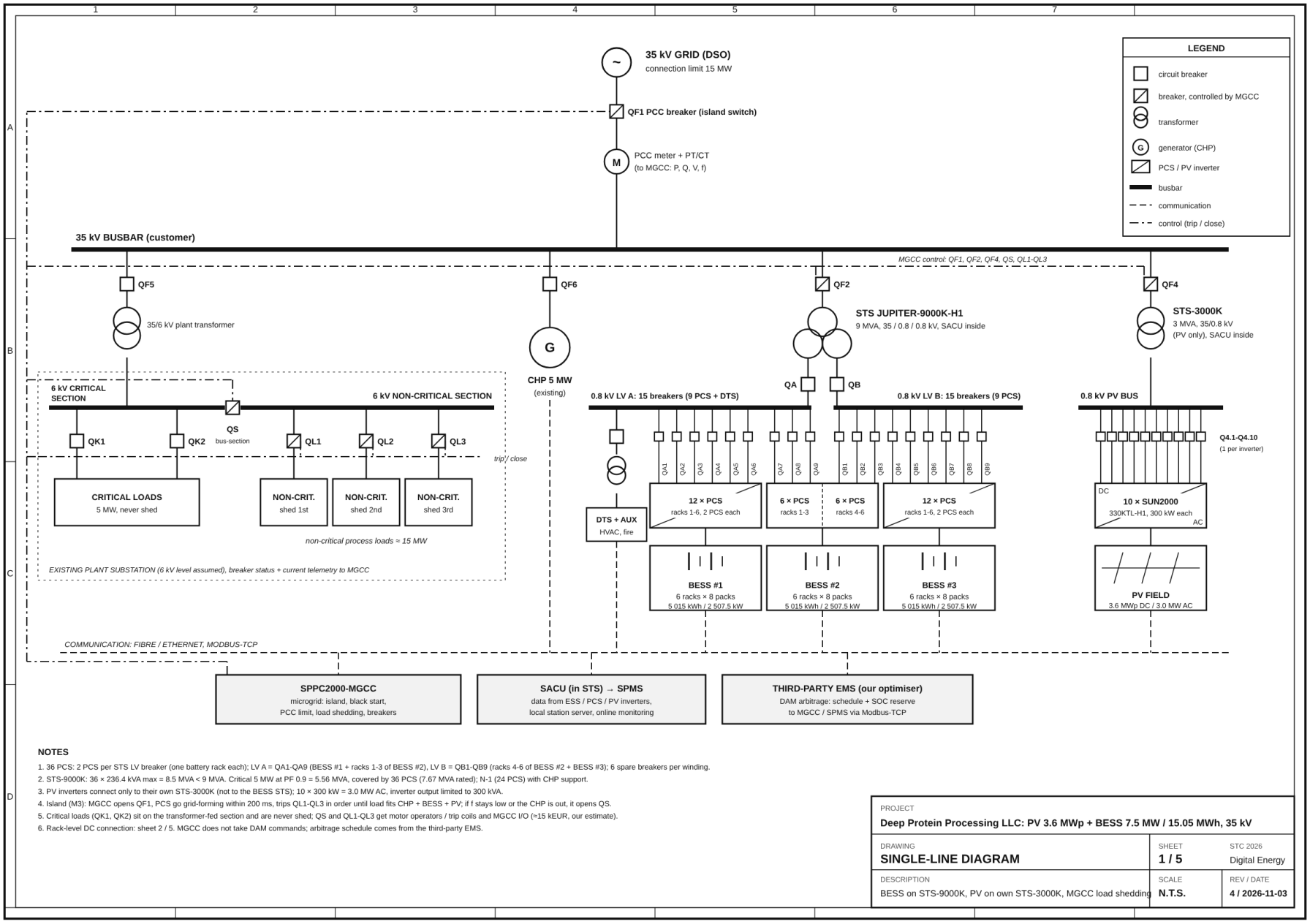
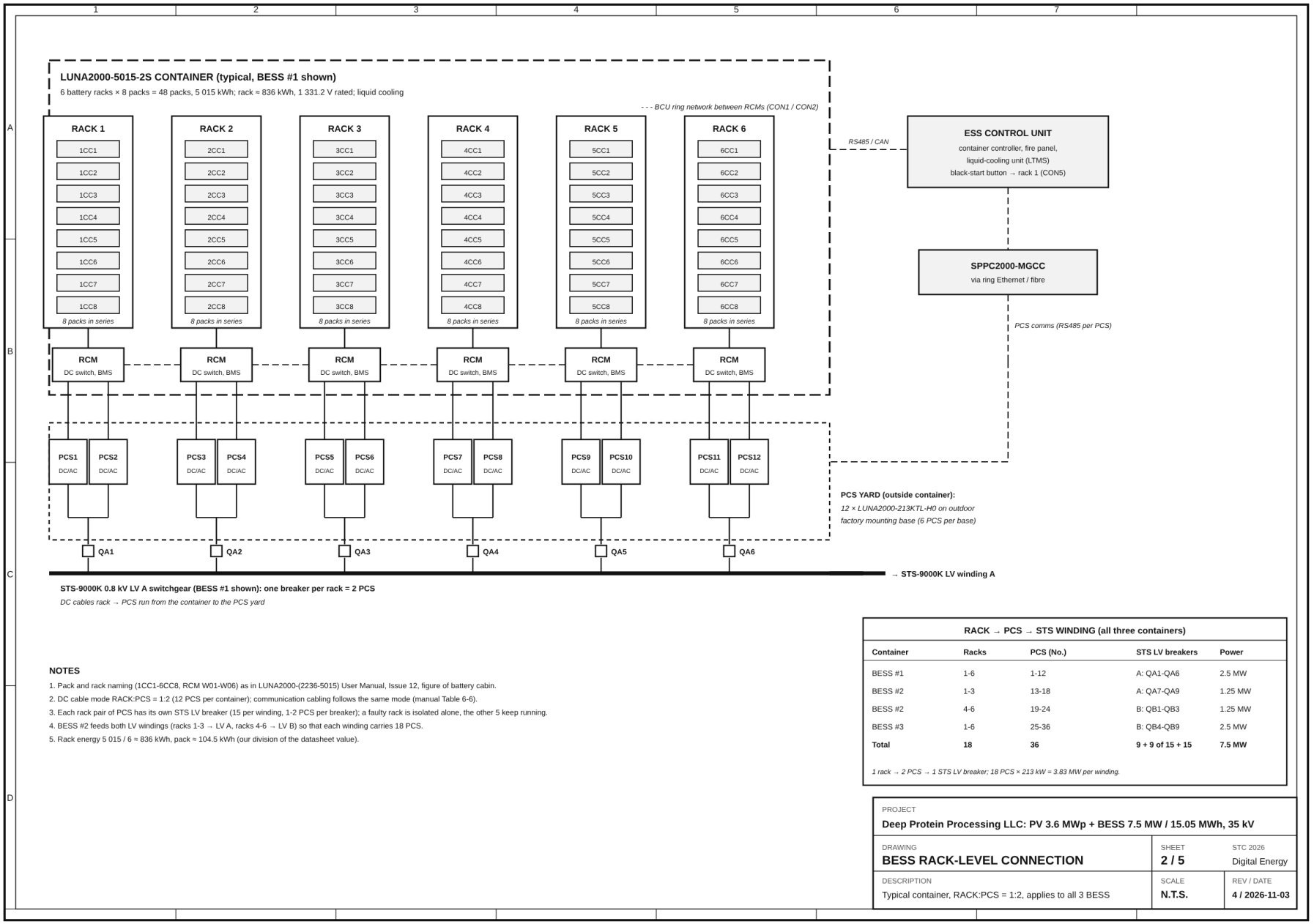


Figure 1. Single-line diagram: BESS on STS-9000K, PV on its own STS-3000K, load shedding by the MGCC



RACK → PCS → STS WINDING (all three containers)

Container	Racks	PCS (No.)	STS LV breakers	Power
BESS #1	1-6	1-12	A: QA1-QA6	2.5 MW
BESS #2	1-3	13-18	A: QA7-QA9	1.25 MW
BESS #2	4-6	19-24	B: QB1-QB3	1.25 MW
BESS #3	1-6	25-36	B: QB4-QB9	2.5 MW
Total	**18**	**36**	**9 + 9 of 15 + 15**	**7.5 MW**

1 rack → 2 PCS → 1 STS LV breaker; 18 PCS × 213 kW = 3.83 MW per winding.

NOTES

1. Pack and rack naming (1CC1-6CC8, RCM W01-W06) as in LUNA2000-(2236-5015) User Manual, Issue 12, figure of battery cabin.
2. DC cable mode RACK:PCS = 1:2 (12 PCS per container); communication cabling follows the same mode (manual Table 6-6).
3. Each rack pair of PCS has its own STS LV breaker (15 per winding, 1-2 PCS per breaker); a faulty rack is isolated alone, the other 5 keep running.
4. BESS #2 feeds both LV windings (racks 1-3 → LV A, racks 4-6 → LV B) so that each winding carries 18 PCS.
5. Rack energy 5 015 / 6 ≈ 836 kWh, pack ≈ 104.5 kWh (our division of the datasheet value).

PROJECT	**Deep Protein Processing LLC: PV 3.6 MWp + BESS 7.5 MW / 15.05 MWh, 35 kV**		
DRAWING	**BESS RACK-LEVEL CONNECTION**	SHEET **2 / 5**	STC 2026 Digital Energy
DESCRIPTION	Typical container, RACK:PCS = 1:2, applies to all 3 BESS	SCALE **N.T.S.**	REV / DATE **4 / 2026-11-03**

Figure 2. Rack-level connection of a container and mapping of all 36 PCS to the STS windings

3.2 Splitting the plant load

The plant cannot be carried in full by its own sources when the grid is lost: the CHP gives 5 MW, the BESS 7.5 MW and the PV plant 0-3 MW, against a load of up to 20 MW. The 6 kV plant bus is therefore split into two sections joined by a bus-section breaker QS:

- **Critical section, 5 MW**. It is fed directly from the 35/6 kV plant transformer and is never shed.
- **Non-critical section, ≈15 MW** in three groups. It is fed through QS.

QS and QL1-QL3 get motor operators or trip coils and are controlled by the SPPC2000-MGCC. Breaker positions and feeder currents go to the MGCC as telemetry, so it always knows how much load is connected. When the grid is lost, the MGCC opens QF1 and trips QL1, then QL2, then QL3 until the remaining load fits the available CHP + BESS + PV power. If the frequency still falls, or if the CHP is out, it opens QS and only the critical section stays on. We assume that the critical feeders can be grouped on one section of the existing switchgear; the actual feeder list of the plant is needed to confirm this.

3.3 Site layout

The three containers stand 4 m apart along their long sides (the forklift clearance in the Huawei manual) and 11.5 m from the STS. The 36 PCS stand on six Huawei factory mounting bases of six PCS each, two bases per container. The fenced area is 29.6 × 20.5 m, i.e. about 607 m². The STS-3000K is placed at the PV field and has its own 35 kV feeder, so it is not inside the BESS fence.

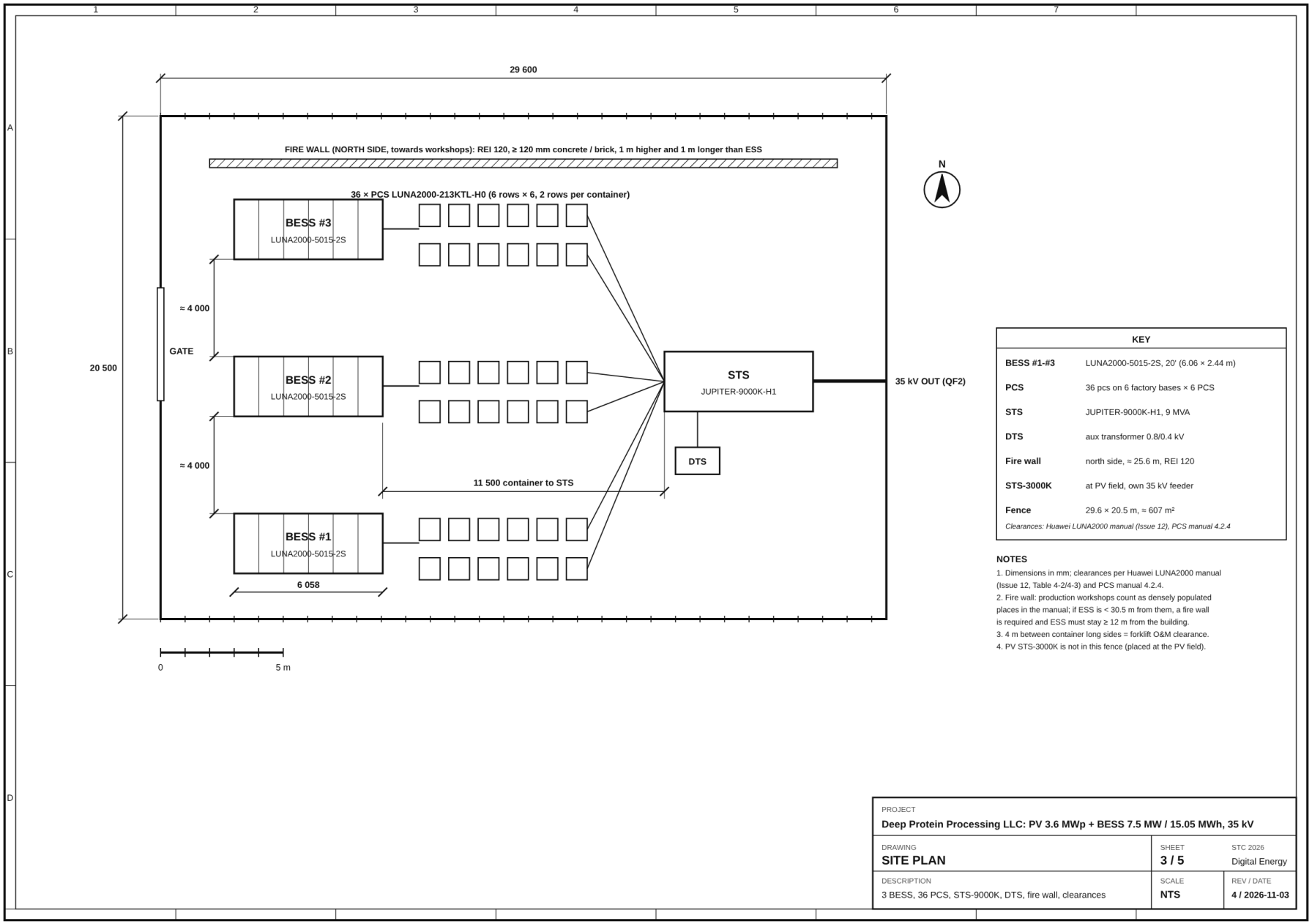


Figure 3. Site plan: three BESS containers, 36 PCS, STS-9000K, DTS and fire wall

3.4 Why the fire wall

The fire wall on the north side faces the production workshops. The LUNA2000 User Manual counts “production workshops of labour-intensive enterprises” as densely populated places and requires more than 30.5 m between the ESS and such a building. If that distance cannot be kept, a fire wall must be built and the ESS must still be at least 12 m from the building. The wall must have a fire resistance of at least 2 hours (for example, reinforced concrete or brick of 120 mm or more) and extend 1 m above and 1 m beyond the ESS. We do not know the real distance to the workshops, so we keep the wall in the design (its cost is part of item 10). If the site survey confirms more than 30.5 m, the wall can be dropped. Toward other buildings, roads and parking places the manual requires at least 3 m, and there a wall must not be used to shorten the distance.

Why the wall is on one side only. The manual asks for a fire wall only where a required distance cannot be kept, and only between the ESS and the object it protects. On the other three sides the rules are already met without a wall:

- toward other buildings, roads and parking places the manual requires at least 3 m and does not allow a wall to replace that distance, so a wall there would add nothing;
- between the containers the manual sets 4 m between long sides and asks for fire walls between ESSs only if these clearances cannot be kept; our layout keeps 4 m, so no walls are needed between the containers;
- the gate side (west) and the STS side (east) must stay open for the forklift, the crane and the fire brigade.

Walls on all four sides would also cost more and would restrict airflow around the liquid-cooling units. If the site survey shows that another side also faces a densely populated building closer than 30.5 m, the wall is extended to that side as well.

4. Operating scenarios

The six scenarios share the roles as follows: arbitrage pays the project back, while peak shaving and island mode protect the plant.

No.	Scenario	What the BESS does	Result
1	Day-ahead market arbitrage	In spring, summer and autumn it charges from PV and in the midday price dip and discharges in the evening peak. In winter it charges at night or during the day and discharges in the evening	About EUR 965 thousand/year at 0% reserve, EUR 878 thousand at 10% reserve
2	Peak shaving above 15 MW	Covers the excess when the CHP is not running or PV output drops	Excess up to 5 MW; three containers hold ≈13.5 MWh usable, i.e. ≈2.7 h at 5 MW
3	Critical load reserve	Holds the SOC reserve and instantly picks up the load if the CHP trips	Two of the three containers already give 5 MW; the CHP remains the main source and the BESS is the bridge
4	Island mode and black start	SPPC2000-MGCC switches to island in ≤200 ms, sheds non-critical feeders; the BESS forms the grid,	Critical 5 MW kept; PV curtailed by frequency (P(f))

No.	Scenario	What the BESS does	Result
		the CHP and PV work in parallel	
5	Ancillary services	Reserves part of the power for frequency regulation	Additional potential only; Ukrenergo rates were not found, so we give no figure
6	Auxiliary consumption	Powers cooling and HVAC, ≈10 kW per container	≈EUR 33 thousand/year, already included

Table 5. Operating scenarios of the BESS

The task contains no load profile, so we built our own: 11 MW at night (00-06), 14 MW between 06 and 08, 17 MW during the day (08-18) with 20 MW peaks at 10-12 and 14-16, then 15 MW in the evening and 12 MW late. Since the CHP provides 5 MW, grid import at the peak is exactly 15 MW.

The main trade-off is the SOC reserve: every 10% of reserve costs about EUR 90 thousand of arbitrage per year (EUR 965 thousand at 0%, 878 at 10%, 789 at 20%, 696 at 30%). We keep a dynamic reserve of 10%, which lasts about 18 minutes at 5 MW. That is enough as a bridge while the CHP is the main backup. We raise the reserve to 30% (about 54 minutes at 5 MW) only when the CHP is out for maintenance or a failure is forecast.

5. Energy management system (EMS)

5.1 Architecture

The EMS is built on two layers.

- **Data collection, SACU and SPMS:** the SACU inside each JUPITER STS collects data from the containers, the PCS and the PV inverters. It passes them to the SPMS, the local station server, which also gives online monitoring over the internet and is connected to the SPPC2000-MGCC.
- **Layer 1, real-time control, SPPC2000-MGCC:** microgrid functions, control of breakers and relays, the 15 MW import guard and load shedding. It works in milliseconds to seconds and always has the last word.
- **Layer 2, third-party EMS, our price optimiser:** our own software on a server at the site. Once a day, after the day-ahead results, it solves a linear programme over 24 hourly prices. It then sends an hourly power schedule and an SOC reserve to the MGCC over Modbus-TCP.

Other signals come from the PCC meter (P, Q, V, f) and the CHP controller. The MGCC acts on breakers QF1 (PCC), QF2 (BESS), QF4 (PV), QS (6 kV bus section) and QL1-QL3 (non-critical feeders).

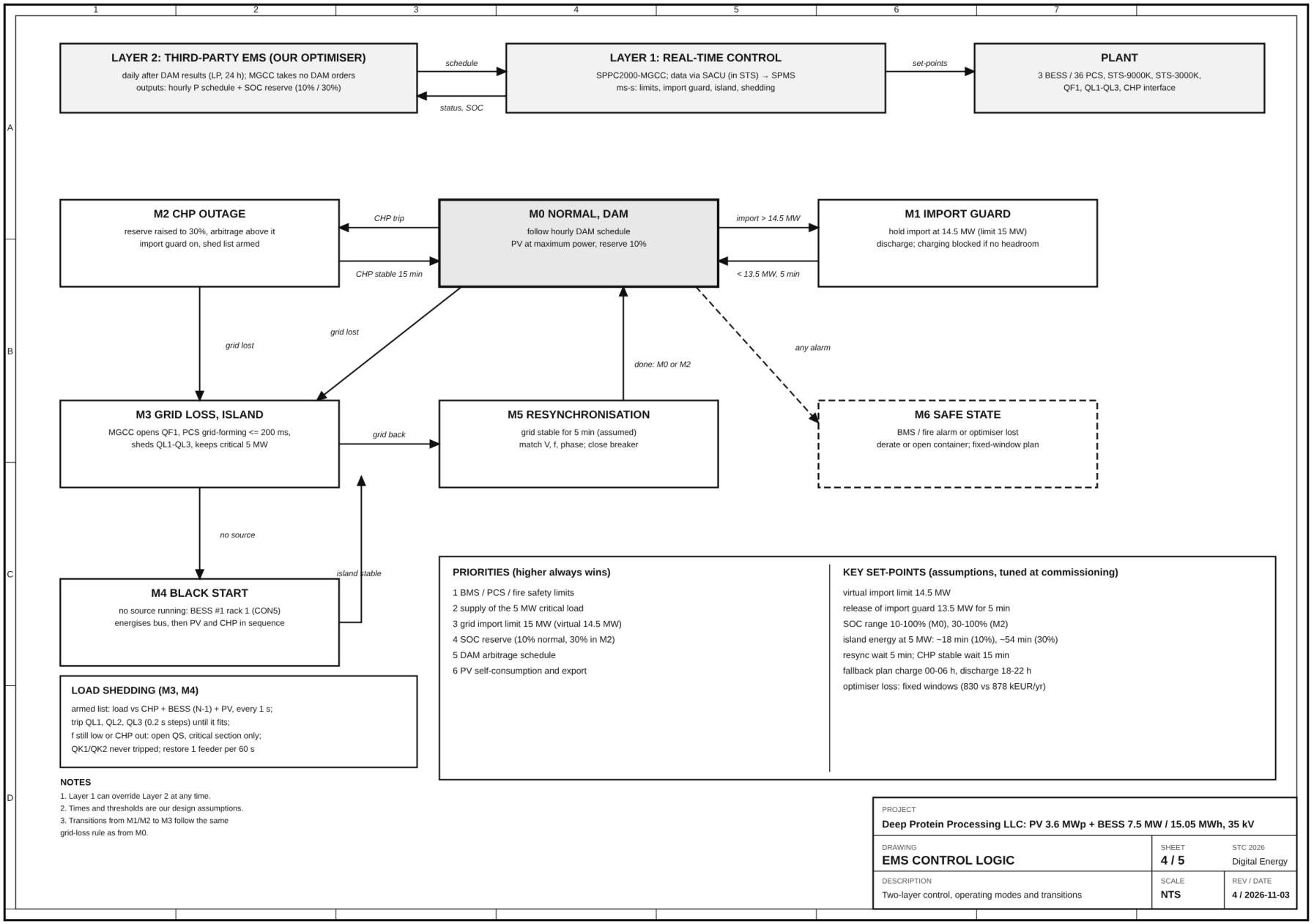


Figure 4. EMS control logic: two control layers, operating modes and load shedding

5.2 EMS options

We compared three EMS options. Because the MGCC does not take market orders, each of them has a third-party layer above it: in option A it is only a seasonal time schedule, in B and C it is our optimiser. Option C is the target, and option A is the minimum acceptable one.

Option	Control logic	Huawei components	Gross BESS margin, EUR thousand/year	Payback, base / stress
A. Fixed windows	Fixed charge and discharge hours by season from a simple scheduler, import guard	SPPC2000-MGCC, SPMS	830	3.6 / 4.6 years
B. Price-based optimisation	A plus the daily optimiser that builds the schedule from day-ahead prices	SPPC2000-MGCC, SPMS, our optimiser	878	3.4 / 4.4 years
C. Microgrid	B plus island mode, black start, load shedding and a dynamic SOC reserve	SPPC2000-MGCC, SPMS, our optimiser	878	3.4 / 4.4 years

Table 6. Comparison of the three EMS options

All values are for three containers, a 10% SOC reserve, a cycle efficiency of 88% and the hard 15 MW import limit. The optimiser adds about EUR 48 thousand per year (≈6%) over fixed windows. Option C adds nothing to revenue, but it protects the 5 MW critical load and meets the on-grid / off-grid requirement. We did not estimate the value of avoided outages.

5.3 Priorities

When goals conflict, a higher priority always wins:

1) BMS / PCS / fire safety limits;
2) supply of the 5 MW critical load;
3) import limit (15 MW, virtual limit 14.5 MW);
4) SOC reserve (10% normal, 30% when the CHP is out);
5) day-ahead arbitrage schedule;
6) PV self-consumption and export.

5.4 Operating modes.

Mode	Entry condition	BESS / PV / CHP action	Exit
M0 Normal, day-ahead market	grid healthy, CHP running	execute the hourly schedule; PV at maximum; 10% reserve	any other entry condition
M1 Import guard	import above 14.5 MW or forecast above it	discharge to hold import at 14.5 MW; charging blocked if the headroom is smaller than the charging power	import below 13.5 MW for 5 min
M2 CHP	CHP trip or maintenance	30% reserve; arbitrage only above	CHP stable for

Mode	Entry condition	BESS / PV / CHP action	Exit
outage	flag	the reserve; import guard active; shed list armed	15 min
M3 Grid loss, island	grid lost (protection signal)	MGCC opens QF1, the PCS switch to grid-forming within 200 ms, non-critical feeders are shed, PV follows the load, the CHP runs in grid-following	grid restored and stable
M4 Black start	island and no source running	rack 1 of BESS1 energises the busbars, PV and the CHP start in turn, loads are reconnected by priority	island stable
M5 Resynchronisation	grid present for a set time	MGCC matches voltage, frequency and phase, closes QF1, returns the BESS to grid-following	M0 or M2
M6 Safe state	BMS fault, fire alarm, loss of the optimiser	the affected container reduces power or disconnects, the other two keep working; if the optimiser is lost, fixed windows take over	alarm cleared by the operator

Table 7. EMS operating modes and transition conditions

5.5 Load management and shedding

Load shedding is what keeps the island stable. Every second the MGCC computes the available power: CHP plus the BESS with one container out plus the current PV output. It keeps an armed list of which non-critical feeders must open if the grid is lost now. When the grid is lost, it opens QF1 and trips the armed feeders in the order QL1, QL2, QL3, so that the remaining load fits the sources. If the frequency keeps falling after the last group, or if the CHP is not running, the MGCC opens the bus-section breaker QS, and the BESS carries only the critical section. The critical feeders QK1 and QK2 are never tripped. In normal operation the same logic prevents an overload when the CHP trips at the peak: the BESS covers the gap and, if needed, QL1 is shed.

For example, at the 20 MW peak with the CHP running, the island sources are 5 + 5.0 + about 1.5 MW of PV, i.e. about 11.5 MW, so QL1 and QL2 open and about 10 MW stay on, including the full 5 MW critical load. Without the CHP the BESS alone carries the critical 5 MW for about 18 minutes at a 10% reserve, or 54 minutes at 30%. After resynchronisation the feeders are reconnected in reverse order, one every 60 s.

5.6 Typical day

Figure 5 shows the optimiser schedule for a median July day. The BESS charges at midday, when the price is lowest and PV is generating, but only within the headroom under 15 MW: around 12:00-15:00 grid import sits exactly at the limit. It discharges in the 20:00-22:00 price peak. The daily margin is about EUR 2 030.

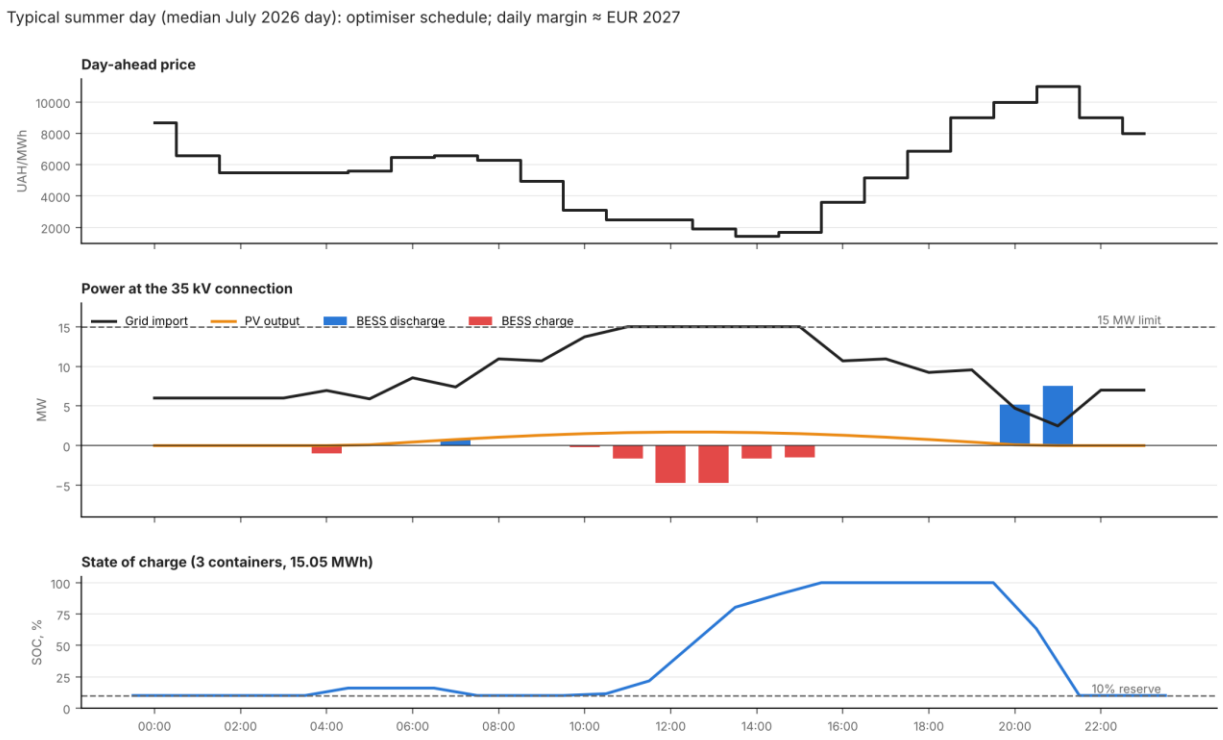


Figure 5. Typical summer day: price, power at the 35 kV connection and SOC

6. Economic feasibility

In the first year the project brings about EUR 1 157 thousand net at a CAPEX of EUR 3.88 million. This gives a simple payback of 3.4 years and an IRR of 28.0%. With a 30% lower BESS margin the payback is 4.4 years.

Item	**EUR thousand/year (year 1)**
BESS arbitrage, option B/C, 10% SOC reserve, 15 MW import limit	878
Container auxiliary consumption	-33
PV energy at day-ahead price (profile-adjusted)	246
Avoided grid tariff on PV energy (≈UAH 1 434/MWh)	116
Total before operation and software	1 207
Operation (O&M ≈32) and optimiser software (10)	−42
Battery capacity fade in the first year	−8
Net benefit, year 1	1 157

Table 8. First-year revenue breakdown

The calculation uses hourly day-ahead prices for January, April, July and October 2026, linear-programming dispatch with the 15 MW import limit, a cycle efficiency of 88%, one cycle per day and an exchange rate of UAH 51.5/EUR. Grid tariffs: transmission 928.45, dispatching 118.64, class 1 distribution 386.60 UAH/MWh. Capacity fade lowers the BESS margin almost linearly, from EUR 870 thousand in year 1 to EUR 621 thousand in year 16.

6.1 Operating costs, NPV and IRR

Assumption	Value	Basis
Operation (O&M)	1.5% of the Huawei equipment CAPEX per year, ≈EUR 32 thousand	Our estimate
Optimiser software	EUR 30 thousand one-off, EUR 10 thousand/year for support	Our estimate for in-house development
Auxiliary consumption	10 kW per container, ≈EUR 33 thousand/year	Conservative estimate
Discount rate	10% (sensitivity 8% and 12%)	Our assumption
Degradation	Linear to 60% capacity in year 20	Huawei 8 200 cycles / 20 years

Table 9. Financial model assumptions

Indicator	Base	Stress (BESS margin × 0.7)
Net benefit, year 1, EUR thousand	1 157	896
Simple payback	3.4 years	4.4 years
Discounted payback (10%)	4.4 years	6.2 years
IRR, 20 years	28.0%	21.2%
NPV at 8% / 10% / 12%, EUR million	6.2 / 5.0 / 3.9	4.0 / 3.0 / 2.2
Cumulative net benefit over 20 years, EUR million	15.8	11.5

Table 10. Base and stress results: payback, IRR and NPV

6.2 Sensitivity

Scenario	Net income year 1, EUR thousand	Payback, years	IRR, %	NPV 10%, EUR million
BESS margin ×0.7 (stress)	896	4.4	21.2	3.02
BESS margin ×0.85	1 027	3.9	24.6	3.99
BESS margin ×1.15	1 288	3.1	31.4	5.94
CAPEX +10%	1 157	3.8	25.3	4.58
CAPEX −10%	1 157	3.1	31.4	5.35
SOC reserve 30% all year (−EUR 182 thousand/year)	976	4.1	23.3	3.60
PV CAPEX EUR 300/kW DC	1 157	3.6	26.4	4.75
PV CAPEX EUR 200/kW DC	1 157	3.2	29.8	5.18
PV generation −10%	1 121	3.5	27.0	4.67

Table 11. Sensitivity analysis

The case stays positive in every row. The BESS without PV is also profitable (CAPEX about EUR 2.56 million, NPV about EUR 3.30 million, IRR about 28.7%), and the PV plant adds about EUR 1.66 million of NPV.

6.3 Tornado diagram

In Figure 6 each input is moved by ±20% while the others stay at base. The project stays profitable in every case: the lowest NPV is EUR 3.38 million and the longest payback 4.2 years. The exchange rate matters most, because all revenue is in hryvnia and all CAPEX in euro, followed by the BESS arbitrage margin and the CAPEX. Degradation and O&M hardly move the result. The discount rate affects only the NPV, so it is not shown on the payback side.

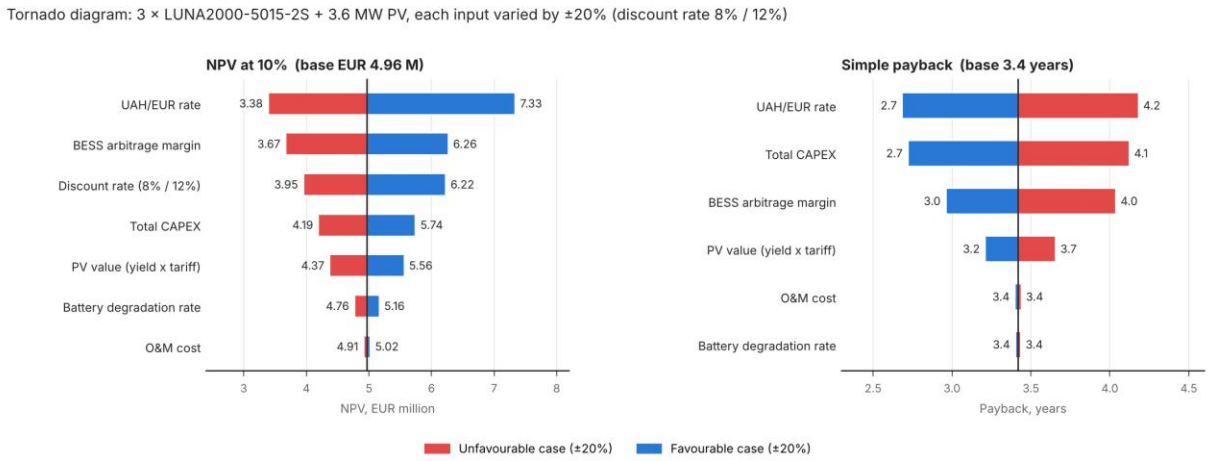


Figure 6. Tornado diagram of NPV and simple payback, each input ±20%

7. Sizing decision and grid import limit

The client's connection limit is 15 MW, the peak consumption is 20 MW and the CHP gives 5 MW, so at the peak import is exactly at the limit. That is why the dispatch model treats the limit as a hard constraint: the BESS charges only within the headroom under 15 MW. In the peak windows 10-12 and 14-16 h this headroom is only 0.2-1.6 MW, at other times 3-4.7 MW. For three containers the limit costs about EUR 16 thousand per year.

Size	CAPEX, EUR million	BESS margin, EUR thousand per year	Net income year 1, EUR thousand	Payback base / stress, years	IRR base / stress, %	NPV 10% base / stress, EUR million	Critical 5 MW with one container out
1 container	2.33	297	621	3.8 / 4.5	25.1 / 21.3	2.54 / 1.88	No (0 MW)
2 containers	3.10	593	894	3.5 / 4.4	27.1 / 21.4	3.78 / 2.47	No (2.5 MW)
3 containers (chosen)	3.88	878	1 157	3.4 / 4.4	28.0 / 21.2	4.96 / 3.02	Yes (5.0 MW)
4 containers	4.78	1 142	1 398	3.5 / 4.6	27.4 / 20.1	5.89 / 3.34	Yes (7.5 MW)

Size	CAPEX, EUR million	BESS margin, EUR thousand per year	Net income year 1, EUR thousand	Payback base / stress, years	IRR base / stress, %	NPV 10% base / stress, EUR million	Critical 5 MW with one container out
5 containers	5.54	1 384	1 618	3.5 / 4.7	27.4 / 19.7	6.82 / 3.71	Yes (10.0 MW)
6 containers	6.32	1 615	1 828	3.5 / 4.8	27.1 / 19.2	7.64 / 4.00	Yes (12.5 MW)

Table 12. Sizing comparison by number of containers

We choose three containers. It is the smallest size that keeps the 5 MW critical load with one container out, and it has the best payback and IRR. More containers keep adding NPV, but only because the model assumes that the BESS does not move market prices. At the same time IRR, stress IRR and stress payback get worse with every container, and the CAPEX the client has to finance grows by 63% at six containers. We see three containers as the balance point: it meets the technical task with security and gives the best return per euro.

Why not more than three:

- **Transformer.** Four containers give 10 MW, more than the 9 MVA of the STS-9000K, so a second BESS transformer and a third 35 kV bay would be needed.
- **Import limit.** The BESS may charge only within the headroom under 15 MW, so each extra container earns less: about EUR 297, 296, 293, 285, 277 and 269 thousand per container per year for one to six containers.
- **Return.** The IRR is highest at three containers and falls to 27.4% at four and 27.1% at six; the stress IRR falls from 21.2% to 19.2%. The extra EUR 0.90 million for a fourth container earns less per euro.
- **Market.** 10 MW / 20 MWh in one area starts to move day-ahead prices, which weakens the price-taker assumption behind all figures.
- **Security and maintenance.** Three containers already cover the 5 MW critical load with one container out, so one container can also be taken out for planned maintenance without losing the critical reserve. A fourth adds no security for the critical load.

Why not fewer: one container gives no security at all, and two give exactly 5.0 MW with no margin and only 2.5 MW if one fails.

8. PV plant capacity and separate transformer

Parameter	Value
PV plant capacity	3.6 MW (DC) / 3.0 MW (AC), DC/AC ratio 1.2
Transformer	STS-3000K, 3 MVA, own 35 kV feeder (QF4)
Annual generation	about 4 157 MWh (1 154.6 kWh/kW, PVGIS-5)
PV CAPEX (our estimate)	about EUR 1 072 thousand (modules and structures 900, inverters 67, STS-3000K 105)
Annual value	EUR 246 thousand at day-ahead price; EUR 362 thousand including the saved grid tariff
Simple payback of the PV plant	3.0 years with the tariff, 4.4 years for energy only (3.6 and 5.2

Parameter	Value
alone	years with 20% contingency)
Share of annual consumption	about 3% (4.2 of ≈131 GWh)
Field area (our estimate)	about 3.6-4.3 ha at 1.0-1.2 ha/MW, no source

Table 13. Key parameters of the PV plant

The PV plant is the cheapest energy the plant can add, and in spring, summer and autumn it charges the BESS in the midday price dip, so PV and BESS raise each other's value. A larger PV plant would need a larger transformer; we keep 3.6 MWp as a conservative size for the available area and budget.

9. Assumptions and risks

The most sensitive financial parameters are the exchange rate, the BESS margin and the CAPEX. The full list of assumptions is in the table.

Topic	Our assumption	Basis or risk
Power factor of the critical load	0.9	Measurements at the plant needed; at PF 0.9 the critical load is 5.56 MVA
STS LV breakers	15 per winding, 1-2 PCS each	As described by the mentors; JUPITER STS manual not in our materials
Cycle efficiency	88% base	Battery efficiency 95% (Huawei declaration, EU Reg. 2023/1542); the AC value with PCS and STS
STS-9000K windings	18 PCS per LV winding, 4.5 MVA per winding	3.83 MW per winding, 7.5 MW total against 9 MVA
STS-3000K	10 inverters × 300 kW = 3.0 MW	Inverter output limited to 300 kVA by the plant controller
Auxiliary consumption	10 kW per container, ≈EUR 33 thousand/year	Conservative estimate
CHP and grid-forming BESS in parallel	The BESS forms the grid, the CHP follows	Confirmation from the CHP control system is required
Load groups	Non-critical ≈15 MW in three feeders of ≈5 MW; critical 5 MW on separate feeders	Actual feeder list of the plant is needed
SOC reserve	Dynamic 10%, 30% when the CHP is not running	10% costs ≈EUR 90 thousand/year
Client load profile	Own profile	
Hourly PV profile	Half-cosine, scaled to monthly PVGIS output	No hourly PVGIS data
PV CAPEX	≈EUR 250/kW	Source: small rooftop PV; range EUR 200-300/kW
Works, 35 kV bays, fire wall, load shedding	≈EUR 188 thousand plus 20% contingency	Our estimate
Ancillary services	Not monetised	
Market data	Day-ahead prices of four months	The October file covers only 1-15

Topic	Our assumption	Basis or risk
	(January, April, July, October 2026) represent the seasons	October; prices may change as storage grows
Densely populated place	Fire wall kept; ESS ≥ 12 m from workshops	Site check required; wall can be dropped at > 30.5 m
Grid tariff netting	Storage operators have netting under Law 2046-IX	Confirm the status for combined PV + BESS
Island and resynchronisation time	Resync after 5 min of stable grid; CHP stable 15 min; island transition ≤200 ms	Our design values; 200 ms from SPPC2000-MGCC data
Implementation period	12 months	Our estimate; depends on grid approval and equipment delivery
Financial model	20 years, 10% discount, O&M 1.5% of Huawei CAPEX, software EUR 30 thousand + 10 thousand/year, degradation to 60%	Price-taker assumption covered by the ×0.7 stress

Table 14. Assumptions, basis and risks

The main risks are as follows. First, the day-ahead spread will compress as more storage enters the market; the ×0.7 stress case covers this. Second, capacity falls by 40% by the end of life; a fourth container or new packs can be added later, but that would need a second BESS transformer. Third, the BESS depends on one STS-9000K; a transformer fault stops the BESS but not the PV plant, which now has its own STS.

Sources: Huawei LUNA2000-(2236-5015) User Manual and Maintenance Manual (Issue 12), LUNA2000-213KTL-H0 PCS User Manual, LUNA2000 Regulatory Compliance Statement (Issue 04, 2026-09-08), PVGIS-5 report (SARAH3) for 49.397° N, 32.018° E, day-ahead and intraday price files and the equipment price list from the organisers, Ukrenergo tariffs (transmission UAH 928.45/MWh per the organisers' analytical note, dispatching UAH 118.64/MWh from 1 July 2026), Cherkasyoblenergo distribution tariff (UAH 386.60/MWh, voltage class 1, from 1 August 2026), public Huawei datasheets for JUPITER-9000K-H1, SUN2000-330KTL-H1 and SPPC2000-MGCC (solar.huawei.com).

10. Implementation schedule

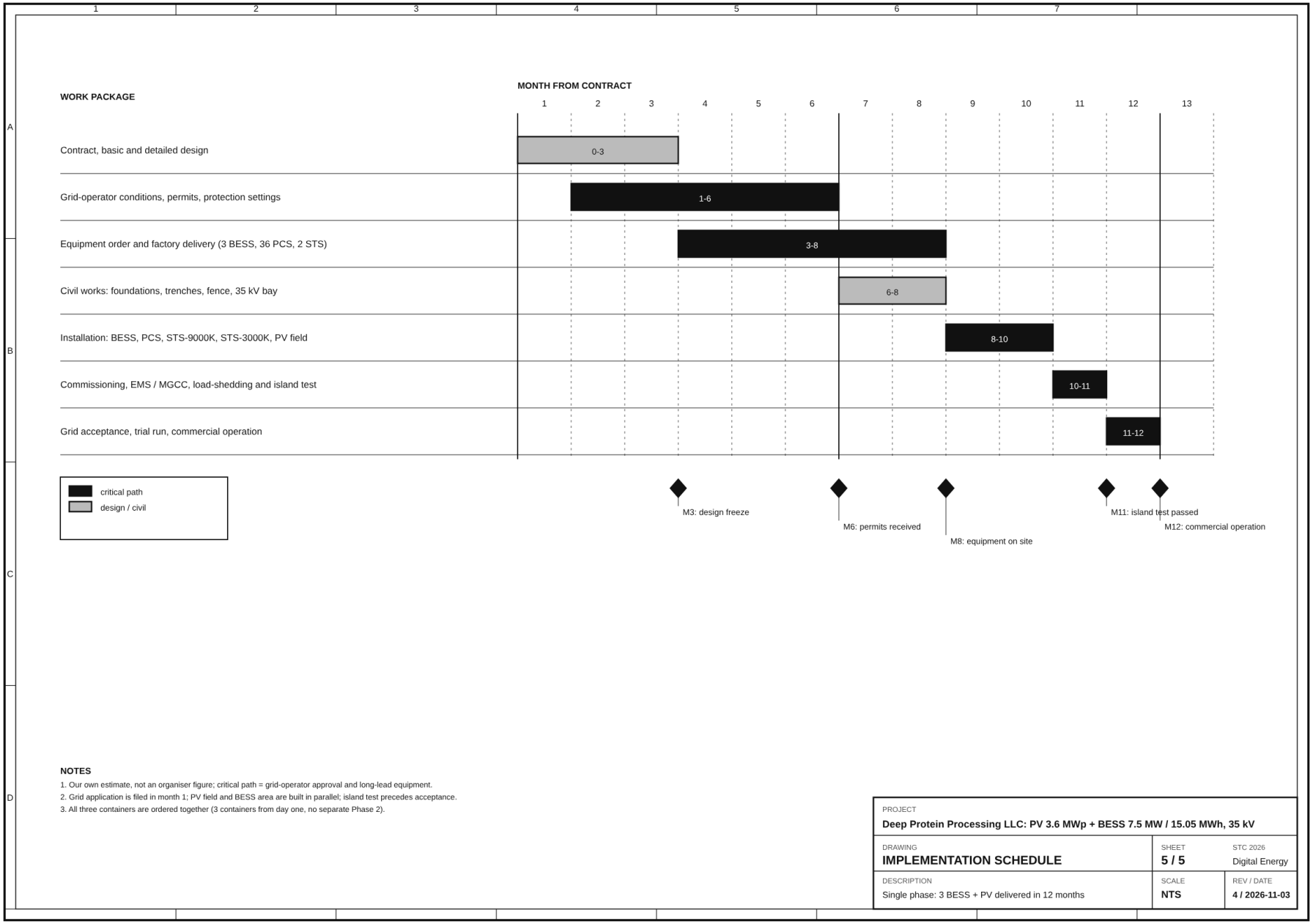
The schedule is our estimate. The critical path runs through grid-connection approval and long-lead equipment. All three containers and both transformer stations are ordered together.

Work package	Start, month	End, month
Contract, basic and detailed design	0	3
Grid operator's technical conditions, permits, protection settings	1	6
Ordering and factory delivery (3 containers, 36 PCS, STS-9000K, STS-3000K, inverters)	3	8
Civil works: foundations, cable trenches, fence, fire wall, two 35 kV bays	6	8
Mechanical and electrical installation, PV field, load-shedding upgrade of feeders	8	10
Cold and hot commissioning, EMS and MGCC tests, load-shedding and island test	10	11

Work package	Start, month	End, month
Grid operator acceptance, trial operation, commercial operation	11	12

Table 15. Implementation schedule (months from contract)

The containers and the STS arrive ready from the factory, so site work comes down to foundations and cables; the PV field and the BESS area are built in parallel; the island and load-shedding test is done before acceptance. The main schedule risks are the duration of permits and equipment delivery, and both are reduced by filing the connection application in month 1.



NOTES

1. Our own estimate, not an organiser figure; critical path = grid-operator approval and long-lead equipment.
2. Grid application is filed in month 1; PV field and BESS area are built in parallel; island test precedes acceptance.
3. All three containers are ordered together (3 containers from day one, no separate Phase 2).

PROJECT	**Deep Protein Processing LLC: PV 3.6 MWp + BESS 7.5 MW / 15.05 MWh, 35 kV**		
DRAWING	**IMPLEMENTATION SCHEDULE**	SHEET **5 / 5**	STC 2026 Digital Energy
DESCRIPTION	Single phase: 3 BESS + PV delivered in 12 months	SCALE **NTS**	REV / DATE **4 / 2026-11-03**

Figure 7. Implementation schedule, months from contract

11. Innovations in the solution

The table lists only what is built into the solution; the last item is a roadmap and is not included in the economics.

No.	Innovation	What is special	Where in the report
1	BESS as virtual grid capacity	The 20 MW peak is served under a 15 MW connection thanks to import-aware dispatch, so no grid reinforcement is needed	Sections 5.4, 7
2	N-1 secure critical supply	Three independent containers: the loss of any one still leaves 5.0 MW for the critical load; one STS breaker per rack isolates a single rack	Sections 2, 3.1
3	Automatic load shedding by the MGCC	Armed shed list updated every second; non-critical feeders open in 0.2 s steps, then the bus-section breaker QS if needed; critical feeders never	Section 5.5
4	Dynamic reserve linked to the CHP	10% in normal operation, 30% when the CHP is out; the reserve is a priced decision, not a fixed rule	Sections 4, 5.4
5	Degradation-aware dispatch	A cycle is made only when the spread exceeds the wear cost: about EUR 227 per cycle against a median daily margin of about EUR 2 170; no sampled day was below the cost	Section 5.6
6	Grid-forming island and black start	Huawei grid-forming PCS and SPPC2000-MGCC; unplanned island transition within 200 ms; black start from rack 1 of BESS1	Sections 5.4, 5.5
7	Joint PV and BESS optimisation	Separate transformers, one controller: the optimiser sees PV, consumption, the CHP and prices together	Sections 3.1, 5.6
8	Multi-layer safety	Room-level gas suppression and optional pack-level suppression, water sprinkler, detectors, suppression panel, liquid cooling, fire wall towards the workshops	Sections 3.4, 13
9	Value-stacking roadmap (not in the figures)	First the day-ahead market, then the intraday market and ancillary services once rates are published; same EMS interface	Section 4

Table 16. Innovations in the solution

12. CAPEX and OPEX optimisation

To keep the client's costs low we took several steps.

- One SPPC2000-MGCC replaces the separate SPPC and microgrid controller (EUR 36 thousand saved).
- The BESS uses one STS-9000K for all 36 PCS; the PV plant uses the smallest STS in the price list (STS-3000K) that fits its 3.0 MW AC.
- Load shedding reuses the plant's existing 6 kV breakers with motor operators instead of new switchgear (≈EUR 15 thousand).
- The DC/AC ratio of the PV plant is 1.2, so the inverters and the STS-3000K work at full load for more hours.
- Huawei equipment is highly efficient: the LUNA2000-213KTL-H0 PCS reaches 99.01%, the battery cycle efficiency is 95% , and we take 88% for the AC system.
- OPEX is O&M at 1.5% of the Huawei CAPEX (about EUR 32 thousand/year), optimiser software (EUR 10 thousand/year after EUR 30 thousand one-off) and auxiliary consumption of 10 kW per container.
- Fallback: if the optimiser fails, fixed windows still give EUR 830 thousand/year against EUR 878 thousand/year.

13. Operation, protection and compliance

- **Operation and maintenance.** The containers are monitored remotely through the Huawei management system. According to the Huawei maintenance manual, the cooling circuit, fans, fire detection and sprinkler, and the tightening torque of connections are checked periodically, and the fire safety of the ESS at least once a month. The cost is included in OPEX (1.5% per year).
- **Protection.** The 35 kV breakers QF1, QF2 and QF4 and the relay protection are agreed with the grid operator. Anti-islanding protection is disabled only in controlled island mode. The STS and PCS protections are set per the Huawei manuals, and the setpoints are fixed at the design stage (months 3-6).
- **Grid code and permits.** The grid operator's technical conditions, the connection of the generation and storage installation and grid-code compliance are part of the schedule in section 10.
- **Safety and standards.** The equipment comes with a Huawei declaration of conformity. The manual requires the fire separation distances or fire walls of local standards (it names GB/T 51048-2025 and NFPA 855); our fire wall and distances are explained in section 3.4.
- **End of life.** The LFP battery is rated for 8 200 cycles / 20 years with a 40% capacity loss at the end. Recycling follows the EU battery regulation and the Huawei take-back procedure (to be confirmed with the supplier).